

STAMATIS CHOUDALAKIS

Athens, Greece | +30 6985793600 | st.xoud@gmail.com | <https://www.linkedin.com/in/stchoud/> | <https://github.com/Stchoud>

SUMMARY

Mathematician with a strong foundation in data management principles seeking a Data Engineer role. With proven of experience in designing, developing, and optimizing robust ETL pipelines and data processing workflows using Python and R. Skilled in identifying and resolving data flow bottlenecks and data quality issues, and in building clean, rigorously tested code. Eager to leverage my strong analytical background and problem-solving skills to contribute to efficient and scalable data infrastructure. Military obligations are fulfilled.

EXPERIENCE

Freelancing

Athens, Greece

Sep 2020 – Present

Provided comprehensive courses in diverse mathematical fields to over 80 undergraduate and postgraduate students. I also guided students through their assignments in Python, R, and SQL, covering aspects of development and data science projects, including data curation, web scraping, statistical analysis, mathematical modelling, and effectively communicated complex technical subjects in an accessible manner.

EDUCATION

PhD in Bioinformatics

Mathematics Research Center, Academy of Athens | School of Medicine, NKUA

Mar 2024 – Present

Thesis: "Cancer driver gene identification using machine learning".

Developed state-of-the-art machine learning and data curation pipelines, identifying 42 previously unreported mutational patterns in cancer initiation. Optimized computational workflows and wrote reproducible code (Python, R, Bash, Docker).

MSc in Applied Mathematics (Grade: 8.11/10)

Department of Mathematics, NKUA

Sep 2020 – Jun 2022

Dissertation: "Network Graphs of Cancer Mutations" in <https://pergamon.lib.uoa.gr/uoal/object/3237319>

Syllabus included advanced topics in statistical simulation, statistics of machine learning, and numerical linear algebra.

BSc in Mathematics (Grade: 6.29/10)

Department of Mathematics, NKUA

Sep 2015 – Sep 2020

PROFESSIONAL CERTIFICATES

IBM Data Engineering

IBM | Coursera

Sep 2025

Professional Certificate in Data Engineering, with coursework in Python for Data Science, Relational Databases & SQL, ETL & Data Pipelines, Data Warehousing, Big Data (PySpark, Spark SQL, Hadoop), Machine Learning, BI Dashboards (IBM Cognos Analytics, Google Looker), NoSQL (MongoDB, Cassandra), and orchestration (Apache Airflow).

Snowflake Data Engineering

Snowflake | Coursera

In Progress

Currently pursuing a Professional Certificate in Data Engineering with Snowflake gaining hands-on expertise in Snowflake for data development, science, and engineering.

TECHNICAL SKILLS

Programming Languages

- Python (ETL, data visualization, machine learning, web scraping)
- R (ETL)
- Bash (task automation, inter-language integration)
- SQL (MySQL, PostgreSQL - actively developing proficiency)

Code is freely accessible upon request:

- https://github.com/Stchoud/patterns_intra_clustering
- https://github.com/Stchoud/multiomic_intra_clustering.

Cloud & Data Platforms

Building foundational knowledge in Snowflake, Databricks, and AWS through coursework.

LECTURES

Lectures on Machine Learning

School of Medicine, NKUA

Mar 2024 – Present

PUBLICATIONS

Choudalakis S., Kastas G. A., & Dikaivos N. (2025). Intra-clustering analysis reveals tissue-specific mutational patterns. Computer Methods and Programs in Biomedicine, 263, 108681. (<https://doi.org/10.1016/j.cmpb.2025.108681>)

Choudalakis S., Mitrouli M., Polychronou A., & Roupas P. (2021). Solving high-dimensional problems in statistical modelling: a comparative study. Mathematics, 9(15), 1806. (<https://doi.org/10.3390/math9151806>)

FOREIGN LANGUAGES

English

Level C2, ECPE, University of Michigan

Jun 2013

Italian

Level B1, National Foreign Language Exam System

Jun 2012